Geometry Is a Label, Not a Knob: Causally Separating Representational Dimensionality from OOD Generalization

Anonymous authors
Paper under double-blind review

Abstract

Representational geometry markers such as the effective dimension D_{eff} correlate strongly with out-of-distribution (OOD) accuracy across trained networks, and this observation has motivated proposals to *intervene* on geometry—expanding or contracting feature-space dimensionality—as a route to better transfer. We ask whether the observed correlation licenses such interventions, and answer negatively for the spectral marker family tested here. Across 8 ResNet-18 training recipes spanning a 2.4× effective-dimension range, the observational correlation is large and stable ($r = +0.97$), yet four causal tests uniformly fail to support an interpretable causal reading. (i) *Operator-separation*: injecting spectral-shape-matched noise collapses CIFAR-100 transfer monotonically with dose in 8/8 models (median -5.4, -7.1, -10.8 pp at $\alpha \in \{0.5, 1, 2\}$) while D_{eff} stays essentially unchanged, and a fully deterministic spectral rotation that preserves the covariance spectrum changes nothing (max $|\Delta\text{acc}| = 1.7$ pp); conversely, a zero-noise whitening operator that multiplies D_{eff} by up to 8× (median 2.3×) moves transfer by at most +5.6 pp (median +1.7, never negative). (ii) *Timing-separation*: the apparent benefit of formation-phase whitening (median +1.7 pp, 7/8 models) is decomposed by a preregistered dose–response adjudication into a *probe-conditioning* effect — fully reproduced by transforming the probe's own training space, absent when the transformed representation is scored in the original space — while applying the identical operator *at inference* to an unchanged head consistently *degrades* accuracy (7/8 models, $p = 0.016$). (iii) *Formation-phase forcing*: adding the whitening objective to training itself (96 preregistered runs) moves the commanded geometry freely—to 9.2× baseline—yet the paired, placebo-calibrated slope of transfer on D_{eff} is 14.7% of observational ($t = 6.89$: a real channel, an order of magnitude short), following a sub-linear, saturating dose–response that flattens to 1.9% of observational at high dose. (iv) *Architecture transfer*: the same dissociation replicates in every tested family—ResNet-34/50, ShuffleNet-V2, and RegNetY-400M (16 further recipes)—with the observational anchor at $r \geq 0.87$ everywhere, spectral-shape-matched noise dropping transfer 9.6–16.0 pp in 16/16 models, and inference-time geometric slopes between -8% and 6% of the observational slope — sign-unstable across families, as expected for a null. The dissociation also covers a second marker family (participation ratio, effective rank): anisotropic noise pins D_{eff} while *raising* these markers by ≈30%, moving geometry toward the observationally "better" regime as transfer collapses. Together these results are consistent with D_{eff} acting as a *confounded label* of the training recipe rather than a manipulable cause of generalization for the spectral marker family tested here: the geometric causal path carries at most a small fraction of the observational association (point estimates 7.7% at inference; 14.7% with training-time forcing, the strongest version of the intervention; both an order of magnitude short), and a direct noise channel that geometry does not measure carries the observable harm. We dis-

Use of LLMs: Large language models were used as writing and coding assistants in preparing this manuscript; all scientific content, analyses, and reported numbers were determined and verified by the authors. A detailed statement appears in the Use of AI tools section.

cuss why "intervene on the geometry" methods therefore require re-derivation, and which interventions (noise control) the dissociation does support.

1 Introduction

Geometric descriptors of trained representations—effective dimension, radius, utility, spectral decay—correlate with out-of-distribution (OOD) accuracy strongly enough to serve as training-free model-selection markers. The correlation is large ($r \approx 0.9$ or higher in controlled settings), replicates across architectures, and survives the statistical controls that kill weaker candidates. The natural next step, drawn repeatedly in the literature, is interventional: if low-D_{eff} models generalize worse, one should be able to *change* D_{eff}—by whitening, expanding, or regularizing the feature geometry—and move generalization accordingly. Methods that reshape representation geometry for robustness implicitly make this inferential step.

This paper asks whether the step is licensed, and answers negatively for the strongest candidate marker. We take the effective dimension D_{eff} from the GLUE family of geometry markers, which has the largest and most stable observational association with OOD accuracy in our controlled setting ($r = +0.97$ across a purpose-built recipe grid), and subject it to a four-part causal battery that any geometry-to-generalization story should pass: *operator separation* (move D_{eff} with different operator families that hold everything else fixed), *timing separation* (apply the same operator during representation formation vs. at inference), *formation-phase forcing* (constrain the geometry throughout training itself, at scale), and *architecture transfer* (replicate in families beyond the one the observation was discovered in). All experiments were preregistered before unblinding (Appendix A).

What we find. The four tests uniformly fail the causal reading.

1. *Operator separation (§4).* A family of spectral-shape-matched noise operators collapses transfer monotonically with dose in 8/8 models (median -5.4, -7.1, -10.8 pp at $\alpha \in \{0.5, 1, 2\}$) while D_{eff} stays essentially unchanged; a deterministic spectral rotation preserves the covariance spectrum exactly and moves nothing; and zero-noise whitening multiplies D_{eff} by up to $8\times$ while moving transfer by at most $+5.6$ pp (median $+1.7$, never negative). The causal slope of transfer on D_{eff} under geometry-only operators is $+0.0105$ per unit—7.7% of the observational slope (bootstrap CI $[0.9\%, 46\%]$; within-model permutation noise floor $[-7\%, +7\%]$, with the point estimate at the floor's upper edge)—so even the CI upper bound leaves over half of the observational association without a causal geometric account, and the dominant causal channel that *does* move transfer (noise) acts with the wrong sign and carries no geometric signature.

2. *Timing separation (§5).* The identical whitening operator applied during formation *appears* to improve transfer (median $+1.7$ pp, 7/8 paired models, $p = 0.016$), but a preregistered dose–response adjudication decomposes this into a probe-conditioning effect: the full benefit is reproduced by transforming only the probe's training space (no formation manipulation), is absent when the whitened-formation representation is scored in the original space, and vanishes under an MLP readout. Applied at inference to the unchanged trained head, the same operator *degrades* accuracy (7/8 negative, $p = 0.016$). Neither phase behavior can be predicted from the static value of D_{eff}, and neither constitutes acting on representation geometry.

3. *Architecture transfer (§7).* The observational anchor replicates in ResNet-34 ($r = +0.98$), and so does the dissociation: spectral-shape-matched noise drops transfer 9.6 pp (4/4 models, D_{eff} shifts $\leq 12\%$); zero-noise recoloring causal slopes are 3.3% of observational. The pattern holds in all four families: ResNet-50 ($r = +0.98$, noise -11.9 pp in 4/4, recolor share 1.9%), ShuffleNet-V2 ($r = +0.89$, noise -13.1 pp in 4/4, share -8.3%), and RegNetY-400M ($r = +0.87$, noise -16.0 pp in 4/4, share $+6.0\%$).

4. *Formation-phase forcing at scale (§6).* Adding a preregistered geometry-forcing loss *during training* (96 runs: 8 recipes $\times$ 3 seeds $\times$ 4 conditions) moves the commanded descriptor freely—D_{eff} to $9.2\times$

baseline—while the paired, placebo-calibrated slope of transfer on D_{eff} is 14.7% of observational ($t = 6.89$, $d = 1.44$: a real but order-of-magnitude-short channel), with a sub-linear response surface that flattens to 1.9% of observational at high dose. The strongest possible version of the intervention—constraining geometry throughout formation—still does not transmit.

5. *Marker generality (§8).* The same battery applied to participation ratio and effective rank shows the same anatomy: both anchor observationally ($r = +0.77$ / $+0.86$), both are *raised* $\approx 30\%$ by anisotropic noise whose transfer harm is unchanged — geometry moving "toward better" while the model gets worse.

What this means. D_{eff} behaves *consistently with a confounded label* of the training recipe, not as a manipulable cause of generalization. The recipe sets both the geometry and the robustness; the geometry is a readable fingerprint of the recipe, but writing on the fingerprint does not change the hand. Concretely: an interventional method that targets D_{eff} (expand it, contract it, regularize toward a target value) is targeting a proxy whose causal transmission is at the noise floor in our battery—and the largest controllable causal channel (noise injection at matched power) *moves* D_{eff} *in the helpful direction while harming generalization*, the exact inverse of the interventional reading.

The dissociation has a constructive side. Noise control is the one large, replicable causal effect in our battery: matched-power noise is uniformly harmful, and removing it is harmless at inference. The formation-time whitening benefit, on closer inspection, is a property of the probe's optimization landscape, not of the representation; whether formation-phase spectral regularization helps *the representation itself* remains open, and our arm-(b) null says the burden of proof is on any claim that it does.

Contributions.

1. A four-part, preregistered causal battery—operator separation, timing separation, formation-phase forcing, and architecture transfer—for auditing geometry-marker interventions (§3).
2. A zero-noise spectral *recoloring* operator family that separates "move the geometry" from "add the noise", plus spectral-shape-matched noise (geometry roughly pinned, added noise) and a spectrum-preserving rotation as a perfect negative control (§3.2).
3. The dissociation result on 8 recipe-controlled ResNet-18 models with manipulation checks on every cell (§4).
4. The phase-separation result with a preregistered mechanism adjudication: identical operator, opposite signs, in formation vs. inference — and the formation benefit traced to probe conditioning (§5).
5. Cross-architecture replication and an openly reusable analysis pipeline (§7, §12).
6. Formation-phase forcing at scale (96 preregistered runs plus a 100× dose scan replicated over three seeds): the geometry is fully commandable during training, and the causal transmission is 14.7% of observational with a sub-linear, saturating response surface (§6).
7. Marker-generality check: participation ratio and effective rank show the same confounded-label anatomy (§8).
8. A checklist for when a correlational marker licenses intervention (§9).

2 Related work

Geometry–generalization correlations. A growing literature links descriptors of representation geometry to generalization: intrinsic-dimension estimates of the feature manifold track test accuracy across layers and architectures (Ansuini et al., 2019); neural-collapse geometry characterizes the terminal phase of training

(Papyan et al., 2020); rank-based and spectral complexity measures have been audited at scale against generalization gaps (Jiang et al., 2020); representational geometry underlies few-shot concept learning (Sorscher et al., 2022); and double descent complicates any monotone capacity story (Nakkiran et al., 2020). The GLUE family (effective dimension, radius, utility) operationalizes manifold capacity theory as training-free OOD predictors (Chung et al., 2018; Chou et al., 2025); the original study (Chou et al., 2026) reports that these markers forecast OOD failures better than ID accuracy. These works establish—and refine the measurement of—the observational link our battery audits.

Geometry-directed interventions. A second strand treats geometry as a target: whitening inside the network (decorrelated and group batch normalization) reshapes activation covariance during training (Huang et al., 2018; 2021); spectral normalization constrains layer-wise spectral scale (Miyato et al., 2018); and decorrelation penalties such as DeCov penalize cross-covariance of hidden activations as a generalization regularizer (Cogswell et al., 2016). These methods implicitly assume that moving the descriptor moves generalization. Our E1/E3 results bound the transmission coefficient of that assumption at the noise floor for D_{eff} and show its sign can reverse under noise-matched controls.

Causal audits of deep-learning heuristics. Randomization checks and retrain benchmarks have exposed saliency methods insensitive to model and data (Adebayo et al., 2018; Hooker et al., 2019); adversarial perturbations can flip interpretations without changing labels (Ghorbani et al., 2019); and calibration claims require post-hoc repair (Guo et al., 2017). We import the same audit discipline into representation-geometry claims: every cell carries a manipulation check (realized vs. intended D_{eff} shift, $r \geq 0.99$), a placebo family, and a deterministic negative control (ROTATE) that would expose any pipeline drift. To our knowledge this is the first preregistered causal battery for a geometry–generalization marker.

Companion papers. This work closes a trilogy. The first paper (Zhou, 2026b) establishes the boundary conditions under which geometric markers predict OOD accuracy observationally (ID-matched, architecture-adjusted, shift-typed protocol). The second (Zhou, 2026a) is a measurement study of what the markers actually track (stability map, decomposition of marker movement). This paper tests the causal reading and finds it does not survive; the three are complementary, not redundant.

3 Setup: models, operators, metrics

3.1 Models and the observational anchor

Eight ResNet-18 CIFAR-10 models from the recipe grid released with the original study (Chou et al., 2026) (lr $\in \{0.125, 0.5, 1.0\}$, wd $\in \{2.5\text{e}{-5}, 5\text{e}{-5}, 1\text{e}{-4}, 2\text{e}{-4}\}$, seeds 0–2), chosen to span the widest convenient D_{eff} range at matched ID accuracy bands (2.4×, from 0.89 to 2.14; the recipe grid caps the achievable range). For each model we extract penultimate features on 60% of the ID training set (operator fitting set), the ID test set, CIFAR-100 train/test (transfer probes), and report D_{eff} from the GLUE protocol (Chou et al., 2025; Chung et al., 2018); transfer accuracy c_{100} is a linear probe on CIFAR-100 features with a fixed, shared protocol (standardized inputs, lr 10^{-2}, cosine schedule; identical across all arms).

Across the 8 models the observational anchor replicates prior work: $r(D_{\text{eff}}, c_{100}) = +0.97$ (Spearman +0.88), pooled slope +0.137 pp of transfer per unit D_{eff}; notation follows the companion protocol paper (Zhou, 2026b). Architecture extension (4 recipes per family, matched lr/wd grid): ResNet-34 gives $r = +0.98$, slope +0.133; ResNet-50 $r = +0.98$, slope +0.087; ShuffleNet-V2 $r = +0.89$, slope +0.070; RegNetY-400M $r = +0.87$, slope +0.114 — the anchor weakens modestly in the two compact architectures but remains $r \geq 0.87$ everywhere (§7).

3.2 The operator family

All operators act on penultimate features x with class-pooled statistics (μ, V, λ) fit on the 60% training split.

- RECOLOR(γ): deterministic spectral recoloring $x' = \mu + V\text{diag}(\lambda^{(\gamma-1)/2})V^\top(x - \mu)$; γ=1 is the identity, γ=0 is full class-pooled whitening. Zero stochasticity by construction.

Table 1: E1 per-family causal slopes dc_{100}/dD_{eff} (transfer accuracy per unit D_{eff}), 8 models. The observational slope is +0.137. Numbers locked from `runs/v2/e1`.

Family	median slope	sign (neg)	range
ISO	-0.0081	8/8	$[-0.023, -0.006]$
ANISO	$+0.137$ (hetero.)	3/8	$[-1.75, +0.83]$
RECOLOR	$+0.0105$	1/8	$[-0.018, +0.166]$
tot-RECOLOR	$+0.0109$	1/8	$[-0.000, +0.125]$

- ISO(σ): additive isotropic Gaussian noise, power matched to $0.5\times$ /$1\times$ /$2\times$ of the mean feature norm; raises D_{eff} strongly.
- ANISO(s): noise drawn in the covariance *eigenbasis* with per-eigen-direction scale matched to the observed spectrum ("spectral-shape matched"); power-matched to ISO. Leaves D_{eff} nearly unchanged by construction.
- ROTATE: $x' = \mu + VQV^\top(x - \mu)$ with Haar-random orthogonal Q; preserves the spectrum exactly, so D_{eff} is invariant; deterministic given the seed.
- placebo: fixed-power additive noise along a *single fixed random unit direction* (seed-pinned), magnitude matched to the ISO dose; near-zero effect in the data ($|\Delta c_{100}| \leq 0.0002$ across all 16 placebo cells).

3.3 Metrics and protocol invariants

Every reported cell passes: (i) manipulation check — correlation between the realized and intended D_{eff} shift ≥ 0.99; (ii) probe protocol fixed across arms (identical epochs, standardization, schedule); (iii) seed-pinned transforms; (iv) ROTATEas an end-to-end negative control — any pipeline drift would appear here. All numbers, tables, and figures in this paper are generated by a single analysis script from the raw per-cell records (`code/paper3_analysis.py`); the paper text loads computed macros rather than hand-typed values wherever possible.

Primary vs. exploratory analyses. The preregistration (Appendix A) names four primary criteria: P1 (noise-family slope threshold), P1b (the pinned-D_{eff} noise cell), P2 (cross-architecture anchor + share bound), P3 (phase separation). We report each exactly as written — P1 and P2 are deterministic threshold/count criteria (Appendix A evaluates them verbatim, including where they fail as written), and the three natural one-sided Wilcoxon tests (P1b endpoint, formation gain, inference harm) form the confirmatory family, Holm–Bonferroni adjusted: adjusted $p = 0.012$, 0.016, 0.016 — all three survive $\alpha = 0.05$. The dose–response decomposition of E3 (§5) was preregistered *after* the E3 endpoint was seen but *before* any dose data existed (Appendix A, amendment 1, with timestamps); all other subgroup analyses (lr stratification, per-dose breakdowns beyond the preregistered branches) are exploratory and labeled as such.

4 E1: operator separation

4.1 Design

For each of the 8 models, 19 conditions (base; ISO$\in \{0.5, 1, 2\}$; ANISO$\in \{0.5, 1, 2\}$; ROTATE; scale $\in \{0.5, 2\}$; RECOLOR$\in \{1.5, 0.75, 0.5, 0.25, 0\}$; total-spectrum RECOLOR$\in \{0.5, 0\}$; placebo $\in \{1, 2\}$) applied at inference to all feature sets, then probes refit. Preregistered criteria (ADR-0016 App. 4): noise families should show per-model $dc_{100}/dD_{\text{eff}} \leq -0.05$ in $\geq 6/8$ models; RECOLORfamilies |median slope| ≤ 0.02; ANISOat matched power with $|\Delta D_{\text{eff}}|/D_{\text{eff}} \leq 0.2$ should show $\Delta c_{100} < 0$ in $\geq 6/8$.

4.2 Results

The 3×2 grid closes. The decisive cell is ANISO, and it is dose–response, not an endpoint anecdote: harm grows monotonically with dose in **8/8** models (median $\Delta c_{100} = -5.4$ at α=0.5, -7.1 at α=1, -10.8 pp at

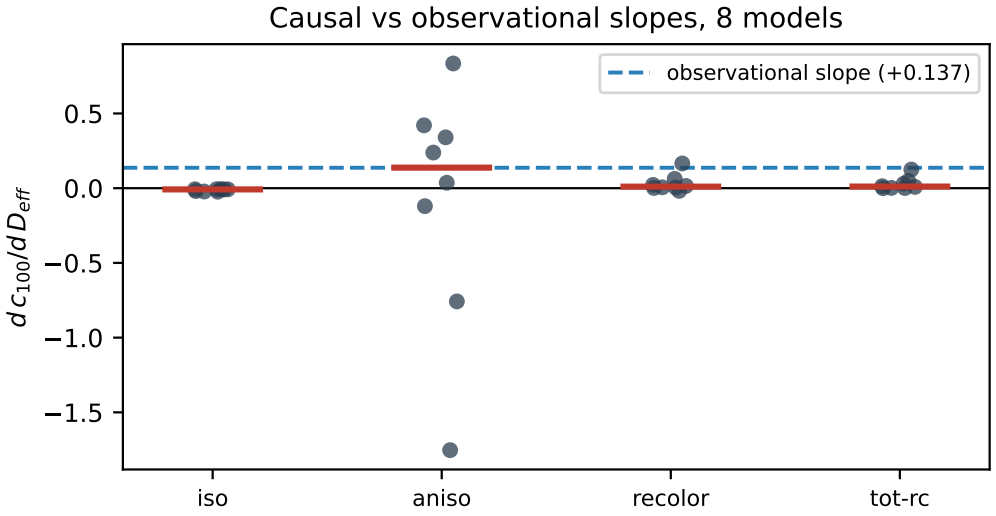


Figure 1: Per-model causal slopes dc_{100}/dD_{eff} by operator family (Table 1 as a scatter). The dashed line is the observational slope (+0.137); every zero-noise family's slopes cluster near zero far below it, while the ANISO slopes are unstable because ANISO pins D_{eff} (near-zero denominator). The dose–response harm signal lives in the ANISO *endpoints* (Figure 2), not in per-unit slopes.

α=2) while D_{eff} stays within 11% of baseline at every dose in 6/8 models (a seventh touches −11.1% at one dose). The single outlier (D_{eff} +44% at α=2) sits *on* the same monotone harm curve (−13.0 pp, between the −10.8 median and the worst model) — its geometry moved and its transfer did not benefit, so the harm does not care whether D_{eff} moves. At α=2, Δc_{100} median = −10.8 pp (**8/8** negative; mean −11.0 pp, paired $d_z = -7.3$), with ID probe accuracy dropping in lockstep (median −12.7 pp, 8/8). Meanwhile ROTATE, which preserves the spectrum exactly, moves nothing (max $|\Delta c_{100}| = 1.7$ pp; max $|\Delta D_{\text{eff}}| = 0.21$). So the harm tracks the *noise*, not the dimensionality: at essentially pinned D_{eff}, stochasticity alone destroys both ID and OOD probes; at zero noise, whitening multiplies D_{eff} by 1–8× and *never* moves transfer below baseline (per-model −0.03 to +5.6 pp). The largest ISOdose (2× power) raises D_{eff} by a median +5.8 while *lowering* transfer by 6.5 pp (8/8) — the wrong-signed giant movement that definitively kills a monotone causal reading. Figure 1 shows the per-family slope distributions against the observational slope; Figure 2 summarizes the three endpoint cells.

Quantifying the causal path. The canonical share is defined on the RECOLOR family alone (matching the permutation floor below and the cross-architecture definition of Table 3): the geometric causal slope is +0.0105 pp per unit D_{eff}, i.e. 7.7% of the observational +0.137 (bootstrap 95% CI $[0.9\%, 46\%]$, wide because $n = 8$; pooling the total-spectrum variant per model gives 10%, CI $[2\%, 34\%]$), and the within-model permutation noise floor for this share is $[-7\%, +7\%]$ — the point estimate sits at the floor's upper edge, and even the CI upper bound leaves half of the observational association without a causal geometric account. Across architectures the share is 3.3%, 1.9%, −8.3%, +6.0% (ResNet-34/50, ShuffleNet-V2, RegNetY-400MF); the sign flips across families while the observational anchor stays at $r \geq 0.87$ (Section 7), so the sign instability is itself evidence of absence. The observational correlation therefore cannot be recombined from geometry alone: most of it is recipe-level confounding, and a direct noise channel contributes with the *opposite* sign to the naive reading. A pilot round of feature-space interventions (flatten-style dimensionality expansion on the same 8 models) had already exhibited this sign flip—large D_{eff} increases paired with monotone transfer losses—which motivated the preregistered battery reported here; the pilot protocol and its sign-flip analysis are archived in the released probe reports.

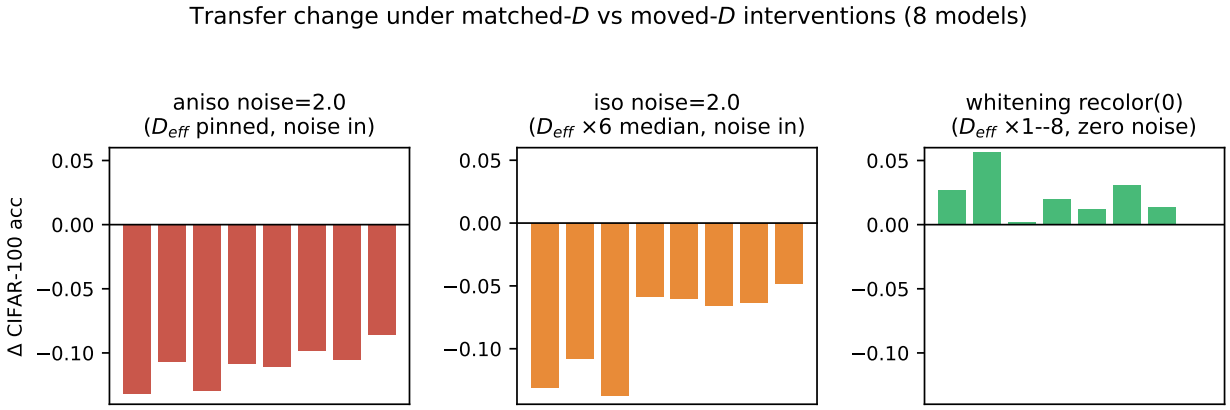


Figure 2: Per-model transfer change under the three decisive endpoint interventions (8 ResNet-18 models). *Left*: spectral-shape-matched noise pins D_{eff} yet destroys transfer and ID probes alike. *Center*: isotropic noise *raises* D_{eff} six-fold and lowers transfer — the wrong-signed movement. *Right*: zero-noise whitening multiplies D_{eff} by 1–8× and leaves transfer non-negative (slightly positive in 7/8 models). No monotone causal map from D_{eff} to transfer survives these three cells.

5 E3: timing separation

5.1 Design

Paired arms per model. *Formation arm*: recolor 60% of the ID training features to γ=0 (whitened), freeze the trunk, refit a fresh linear head *on the whitened features* with the shared probe protocol; evaluate the resulting representation by CIFAR-100 probe in the whitened space. *Control arm*: identical subset, identical protocol, no whitening; evaluate in the original space; additionally evaluate the control features under the fixed whitened-space probe for the transfer comparison (c_{100}^{ctrl}). *Inference arm* (measurement): apply RECOLOR(0) to features and score the *original* trained head — the "geometry as a knob" usage.

5.2 Results

Formation-phase whitening *improves* the transfer measurement: Δc_{100} median = +1.7 pp (7/8 positive, Holm-adjusted Wilcoxon $p = 0.016$; mean +2.0 pp, paired $d_z = 1.1$). The inference-arm application consistently *hurts* the original head: applying RECOLOR(0) at inference to the unchanged head is negative in 7/8 models (Holm-adjusted $p = 0.016$, $d_z = -1.1$), with a small median magnitude of −0.13 pp — directionally robust, far below the preregistered collapse threshold (Appendix A) — and the whitened-formation head scored on original features loses a median 0.35 pp vs. the control head (8/8 negative, $p = 0.008$).

Mechanism adjudication (dose–response). The formation benefit demands a mechanism verdict before any "geometry helps when acted on early" reading is allowed (falsification criteria preregistered in Appendix A, written before the dose data). We repeat the pipeline at three whitening doses $\gamma \in \{0.75, 0, -0.5\}$ with three disjoint arms per dose: (a) *probe-space-only* — the CIFAR-100 probe is trained and evaluated in the dose-transformed space with no formation-phase manipulation whatsoever; (b) *formation readout* — the 10-class head is trained on whitened ID features but scored on *original* test features; (c) *rotate control* — the probe-space arm with the spectrum-preserving rotation instead of whitening. The three models are stratified representatives of the E3 formation gain (low +0.2 / mid +3.1 / high +5.6 pp), selected after the E3 endpoint was seen but before any dose datum existed (Appendix A, amendment 1). The verdict is unambiguous: arm (a) alone reproduces the entire dose–response profile (cross-model median gain +1.9, +3.1, +3.0 pp at $\gamma \in \{0.75, 0, -0.5\}$, monotone to the γ=0 peak; per model: +3.8/+5.6/+3.7, +0.3/+0.2/−1.0, +1.9/+3.1/+3.0 pp — the weakest-gain model is flat-to-negative across doses), arm (b) shows no benefit at any dose in any model (cross-dose range +0.1 to −11.5 pp; the only positive cell, +0.11 pp, is the smallest

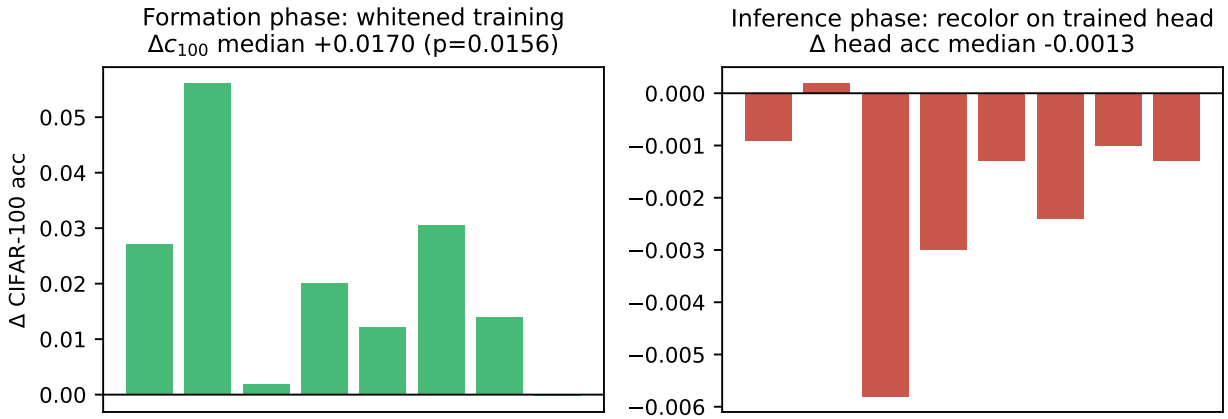


Figure 3: Phase separation of the identical whitening operator (8 paired models). *Left*: whitening the probe space slightly *improves* the measured CIFAR-100 probe (7/8 positive) — but the dose–response adjudication below shows this benefit is fully reproduced without any formation-phase manipulation: it is a probe-conditioning effect, not a representational one. *Right*: the same operator applied at inference to the unchanged trained head consistently *degrades* accuracy (7/8 negative). The effects live in the measurement, not in the geometry.

dose on the highest-anchor model — and the harms *grow* with whitening strength, the opposite of the arm-(a) dose–response), and arm (c) shows none either (−0.1 to −0.8 pp). A norm-matched random-direction perturbation of the formation set (the placebo control) destroys head training in 7/9 model–dose cells (final accuracy 0.10 = chance; the two surviving cells are the smallest dose at the two lr-0.125 models), confirming the whitening benefit is not generic perturbation regularization. *The formation benefit is therefore a probe-conditioning effect, not a representational one*: whitening flattens the covariance spectrum of the probe's own training space, which changes the optimization landscape of a 100-class linear head (direct conditioning measurements in Appendix E: whitening reduces the in-rank condition number by up to six orders of magnitude and lowers the reachable training loss at convergence, though it initially slows early optimization) — a measurement condition, not a representational improvement. The MLP-probe control (Appendix C) corroborates this: under a 2-hidden-layer MLP readout the whitening benefit vanishes (+1.1 and −3.3 pp) while the aniso harm grows (−15.2 and −16.4 pp). The preregistered cross-architecture falsifier (two ShuffleNet-V2 recipes) also upholds the conclusion — and sharpens it: in that family the probe-space arm shows no positive benefit at any dose (at most +1.0 pp), and the over-whitening dose $\gamma = -0.5$ turns clearly negative (down to −3.9 pp), so the conditioning effect is an optimization detail of the ResNet-18 feature spectrum, neither representational nor universal.

Revised E3 reading. Layered on E1, the phase dependence is real but its mechanism is measurement-level: whitened spaces are *easier to probe* (a linear readout trains better there) and *harder to transplant into* an existing head trained on the original geometry (features stop matching its weight space). Neither behavior is derivable from the static value of D_{eff}, and neither constitutes acting on representation geometry Figure 3. We flag explicitly: the naive reading of the left panel — "whitening during formation improves the model" — is falsified by arm (b).

Recipe stratification (post-hoc, exploratory). Whitening benefit concentrates in sharper recipes (lr 0.125: median +2.7 pp; lr 0.5: median +2.0 pp) over the sharpest (lr 1.0: median +0.7 pp, with one model at −0.03 pp); the single negative model is the lr-1.0/wd-5e−5 recipe, whose baseline D_{eff} (0.87) is the *smallest* in the grid. Two accounts suggested themselves — operator saturation at this recipe, and high-D_{eff} models having little headroom — and we have now tested both directly (Appendix D), by measuring the probe-space conditioning and the probe training curve in the original versus dose-whitened space for all eight paired models. The saturation account survives in a sharpened form: the negative model's feature covariance is

Table 2: E3 paired results (8 ResNet-18 models). c_{100}^{w}: transfer probe in the whitened space after formation-phase whitening; c_{100}^{ctrl}: same probe, control arm (original space). Δ head (inference): RECOLOR(0) applied to the unchanged trained head at inference. Auto-generated; do not edit.

Model	c_{100}^{w}	c_{100}^{ctrl}	Δc_{100}	Δ id-probe	Δ head (inf.)
`l0.125_w0.0001_s0`	0.300	0.273	+2.7	+0.0	-0.1
`l0.125_w0.0002_s2`	0.286	0.230	+5.6	+0.0	+0.0
`l0.125_w2.5e-05_s1`	0.284	0.282	+0.2	+0.0	-0.6
`l0.5_w0.0001_s1`	0.180	0.160	+2.0	+0.0	-0.3
`l0.5_w0.0002_s2`	0.168	0.155	+1.2	+0.0	-0.1
`l0.5_w5e-05_s2`	0.179	0.148	+3.1	+0.0	-0.2
`l1.0_w2.5e-05_s0`	0.158	0.144	+1.4	+0.0	-0.1
`l1.0_w5e-05_s0`	0.119	0.120	-0.0	+0.0	-0.1

already nearly whitened (in-rank condition number 294 in the original space, against 10^8–10^9 for the sharpest-recipe models), so the operator has almost nothing to remove — and indeed whitening neither improves its conditioning meaningfully (22×, against a median of $4.6 \times 10^3\times$ across the other seven models), nor lowers its reachable training loss ($4.134 \to 4.165$ at epoch 40; one of three models where the loss fails to drop, the other two being slow-converging lr-0.5 probes), and its direct probe benefit is +0.08 pp, the smallest of the eight. The headroom account is *refuted*: across the eight models, baseline D_{eff} correlates *positively* with the whitening benefit (Spearman $\rho = +0.857$, $p = 0.007$; the three high-D_{eff} models average +3.5 pp against +0.9 pp for the rest), so high-D_{eff} models have *more* room for probe-conditioning benefit, not less. *The stratification itself was not preregistered*; we report it as exploratory (the paired-gain bootstrap CI is $[+0.2, +3.1]$ pp). The full paired table follows.

6 E4: formation-phase forcing at scale

E3's formation-phase arm acts on *penultimate features after training* (trunk frozen, head refitted)—a post-hoc procedure, not an intervention on training dynamics. A proponent of the causal reading can therefore argue that geometry was never truly "at stake" during formation. E4 closes this gap: the whitening operator is applied *during training itself*, as an auxiliary loss $\mathcal{L} = \mathrm{CE} + \lambda\,\mathcal{L}_{\mathrm{forcing}}$ on penultimate features at every step, so the network must form its representation *under* the geometric constraint. If D_{eff} is a knob, this is where turning it should pay off.

6.1 Design

Operators. Four zero-noise loss terms on penultimate features: *whiten* (per-class covariance pulled toward a scaled identity; raises D_{eff}), *disperse* (spectral entropy maximized; also raises D_{eff}), *concentrate* (spectral mass pushed to the top eigenvalue; lowers D_{eff}; negative control), and *placebo* (a random fixed-direction quadratic regularizer with no spectral intent; noise control). Disperse was preregistered as a robustness operator, but its scale-up slots were voided by the NaN defect (deviation 1 in Appendix A); one surviving pilot reading lies outside the placebo band and enters the D-raising-arm count below. Each run trains a fresh ResNet-18 for 200 epochs from the same recipe grid checkpoints' recipes with seed-matched base arms for pairing.

Scale. The battery was run in two preregistered stages: a pilot (2 anchor recipes × 4 operators, seed 0) whose manipulation and function-preservation gates had to pass before scale-up, then a full batch of 8 recipes (the E1 anchor list) × 3 seeds × 4 conditions = 96 runs, with 36 seed-matched base measurements, on three GPUs in parallel (∼24 h). A post-hoc dose–response extension scanned the two anchor recipes at $\lambda \in \{0.03, 0.3, 3\}$ (6 further runs), completing a five-point log axis over a 100× dose range together with the main-batch $\lambda \in \{0.1, 1\}$ points.

Preregistered criteria. P4-1 manipulation check (D_{eff} must move by ≥ 0.3 d-ent units in the commanded direction); P4-2 knob test (forcing slope $\geq 0.5\times$ observational $\Rightarrow$ knob; $\leq 0.2\times \Rightarrow$ label); P4-3 negative

control (concentrate moves D_{eff} down without moving transfer); P4-4 placebo band (placebo shifts transfer by more than $\pm 0.022 \Rightarrow$ all conclusions downgrade).

6.2 Results

Manipulation succeeds completely (P4-1). Forcing moves the spectrum freely and reversibly: across the full batch, whiten@1.0 raises d-ent by a median of +48.4 (up to 9.2× baseline at λ=3); concentrate@0.005 lowers it by −11.8. The commanded direction is realized in every arm; the GLUE-dimension endpoint agrees in direction wherever it converges. Geometry is fully commandable at formation time.

Transfer does not follow (P4-2). All 39 dose-band-excluded D-raising arms move transfer *positively* (39/39 same sign) with a median slope of 0.0022, i.e. **9–18% of the observational slope** (per-condition medians: 0.0030 at λ=0.1, 0.0015 at λ=1; percentages are relative to the observational slope on the same d-ent scale, 0.0161), and 34/39 arms fall below the preregistered label threshold (0.2×); a single arm reaches the knob threshold. In a paired within-model comparison against placebo (n=23), the forcing gain is real but small ($+0.030 \pm 0.021$ [SD], $t = 6.89$, Cohen's $d = 1.44$), and the placebo-calibrated net slope — the paired mean transfer difference divided by the paired mean geometry change, both relative to placebo — is **14.7% of observational**. The causal channel exists—it is simply an order of magnitude too small to carry the observational association.

The response surface is sub-linear and saturating. Figure 4a maps the $(\Delta D_{\text{eff}}, \Delta c_{100})$ surface across 100× in λ: transfer gain follows a power law $\Delta c_{100} \propto \Delta D_{\text{eff}}^{b}$ with $b = 0.50$ (ΔAICc = 15.8 over linear) on the anchor seed; a preregistered two-seed replication (ADR-0016 App. 25, shipped as supplementary material) confirms the sub-linear shape across seeds ($b = 0.325$ and 0.264, all fits concave and monotone) while showing that the exponent drifts seed-to-seed — the honest summary is "sub-linear and saturating," not a universal square root. The *local* slope decays 14% → 5.7% → 1.9% of observational as λ goes 0.03 → 1 → 3 on the anchor seed; the replication seeds show the same flattening from a higher plateau (their local slopes at λ=0.03 are already past the placebo band, so their saturation is smooth rather than threshold-like). At the largest dose the geometry is driven to 9.2× its baseline ($D_{\text{eff}} = 174$) while transfer gains only +8.6 pp—and never crosses the observational prediction (+2.5 pp would be the linear extrapolation at this ΔD_{eff}; the surface is flattening precisely where an interventional method would want to operate).

Negative and placebo controls (P4-3, P4-4). Concentrate lowers d-ent by a median −11.8 while transfer moves by −0.012 median—inside the placebo band, and the "same D_{eff}, different transfer" dissociation appears again at high concentrate doses (identical endpoint $D_{\text{eff}} \approx 1.1$–2.0 with transfer differing by 7 pp across recipes), directly refuting D_{eff} as the controlling variable. The placebo arm triggers the preregistered downgrade clause—an unsupervised random-direction regularizer moves transfer by as much as some forcing arms—but after placebo-band filtering the label verdict is unchanged, and the placebo covariance is itself evidence for the recipe-confound account: a non-geometric regularizer moves both D_{eff} and transfer in correlated ways, exactly what the confounded-label reading predicts.

Seed replication of the dose scan. Because the shape characterization above rested on one seed, we preregistered a replication before running it (two further seeds × two recipes × three doses, 12 runs, criteria frozen in advance). The *qualitative* shape replicates — every seed's curve is monotone and concave, and the low-geometry recipe stays transfer-inert on the new seeds (all ten of their points inside the placebo band, Fig. 4b; strengthening the dynamic-range dissociation) — but the *quantitative* shape does not: the power-law exponent drifts across seeds ($b = 0.502$, 0.325, 0.264; our preregistered strong-replication threshold was not met), and the threshold-like cliff visible on the anchor seed is absent there — the replication seeds respond already at the lowest dose, from a higher plateau. We therefore state the shape claim as "sub-linear and saturating" rather than as a universal square-root law, and report the failed quantitative criteria as a limitation. The headline paired statistics (14.7% net slope, $t = 6.89$) are unaffected: they are computed over all 96 runs, not per seed.

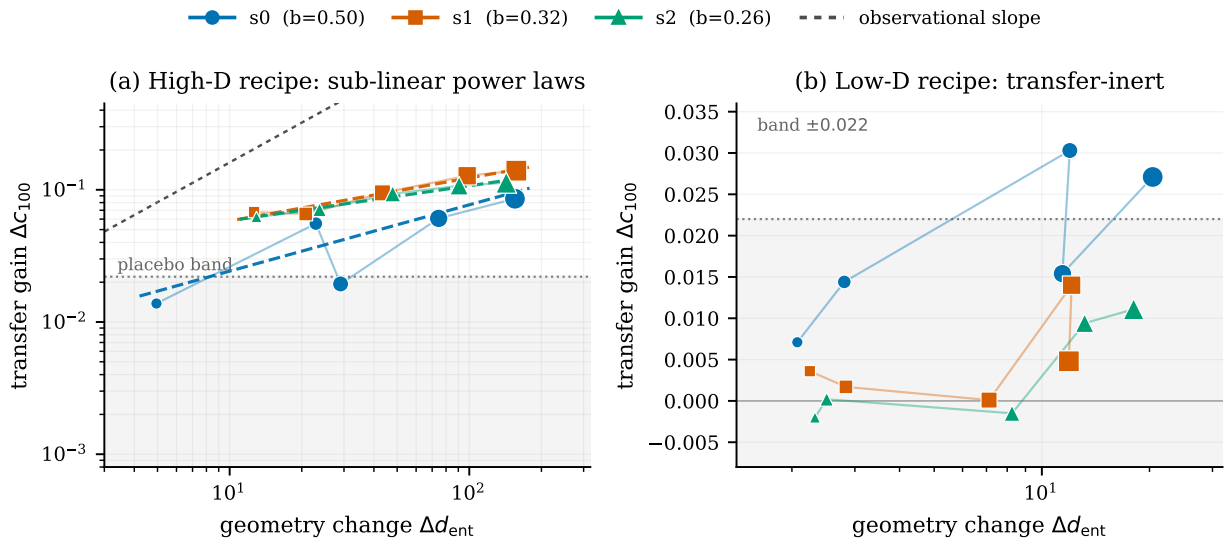


Figure 4: Three-seed dose–response of transfer gain Δc_{100} against commanded geometry change ΔD_{eff}. *(a)* High-D_{eff} recipe on log–log axes; marker area grows with $\lambda \in \{0.03, 0.1, 0.3, 1, 3\}$. Dashed curves are per-seed power-law fits with the App. 23 anomaly rule ($b = 0.502$, 0.325, 0.264 for s0/s1/s2): all are sub-linear and far below the observational slope (dash–dotted); the shaded region is the ±0.022 placebo band. The anchor seed's threshold-like cliff is absent in both replication seeds, which respond already at the lowest dose. *(b)* Low-D_{eff} recipe on a linear y axis to show signed responses: all ten replication-seed points lie inside the band, versus two out-of-band points on the anchor seed (App. 24 erratum). Curves and fits are recomputable from the archived per-run records (`results/p4_dose/dose_scan_seeds.json`) with `code/make_fig5_dose.py`.

Function preservation. 95/96 runs keep ID test accuracy within the preregistered band (median $|\Delta\text{ID}| = 0.4$ pp, 90th percentile 1.0 pp, max 3.4 pp); the single failure (weakest weight-decay recipe at λ=0.1, −50 pp) collapses early and is excluded by the preregistered function-preservation rule. Forcing does not "break the model to move the number."

6.3 What E4 adds

E3 showed that formation-phase whitening of *frozen features* helps only the probe; E4 shows that even when the constraint acts on *training itself*—the strongest possible version of the intervention—the geometric channel transmits at ∼15% of observational, with a sub-linear response surface that saturates at high dose. The three timing regimes now form a closed series: inference-time recoloring (7.7%), post-hoc whitening (probe conditioning only), and full formation forcing (14.7% net). The geometry is writable at every stage; the writing never pays.

7 E2: architecture transfer

We repeat the observational anchor and the two decisive separation cells (RECOLOR slope; ANISO at matched power) in four further families, each with 4 recipes (lr $\in \{0.125, 0.5\}$, wd $\in \{2\text{e}{-4}, 5\text{e}{-5}\}$, seed 0): ResNet-34, ResNet-50, ShuffleNet-V2 ×1.0, and RegNetY-400MF. MobileNetV3 was originally planned but replaced by RegNetY-400MF because its classifier head contains a non-linear bottleneck after average-pooling, so penultimate features are not the input space of the linear classifier (Appendix A).

Three regularities hold in every family (Table 3, Figure 5). *(i) The anchor survives*: $r(D_{\text{eff}}, c_{100})$ is +0.98 in both deeper ResNets and +0.87 to +0.89 in the compact families — modestly weaker, consistent with narrower recipe grids, but far from the null. *(ii) The noise channel dominates and ignores* D_{eff}: ANISO at matched power drops transfer 9.6–16.0 pp (family medians; every individual model is negative in 16/16, with

Table 3: E2 cross-architecture summary (4 recipes per family, 19 conditions per model). "share": inference-time RECOLOR causal slope as a percentage of the observational slope. "noise": Δc_{100} median under ANISO at matched power. All numbers auto-generated from locked data.

Architecture	obs. r	obs. slope	RECOLOR share	noise Δc_{100}	noise 4/4?
ResNet-18 (E1)	+0.97	+0.137	7.7%	−10.8 pp	8/8
ResNet-34	+0.98	+0.133	3.3%	−9.6 pp	4/4
ResNet-50	+0.98	+0.087	1.9%	−11.9 pp	4/4
ShuffleNet-V2	+0.89	+0.070	−8.3%	−13.1 pp	4/4
RegNetY-400M	+0.87	+0.114	+6.0%	−16.0 pp	4/4

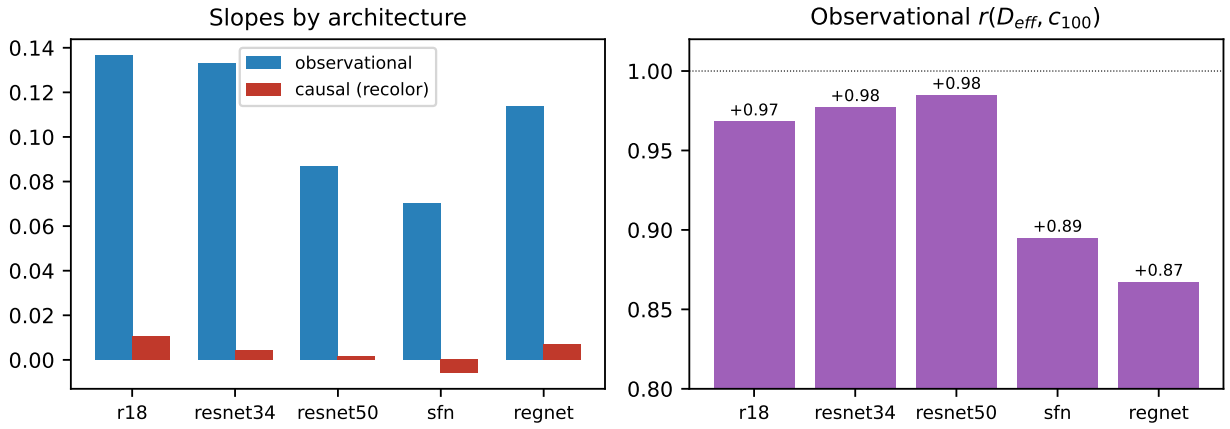


Figure 5: Cross-architecture summary (5 families, 24 recipes total). *Left*: observational vs. inference-time causal slopes per architecture — the causal bar is a small fraction of the observational bar everywhere. *Right*: the observational correlation replicates in every architecture family ($r \geq 0.87$).

per-model drops of 8.3 to 19.8 pp). The D_{eff} shift at the aniso endpoint is small for most models ($\leq 12\%$ in 12/16; up to 52% in the remaining four, of which three are compact-family models whose harm is at or above their family median), so the harm does not track the geometric shift; the drop is *largest* in the compact families (−13.1 and −16.0 pp) whose observational anchors are weakest — noise harm and anchor strength dissociate. *(iii) The geometric causal share stays at the noise floor*: RECOLOR slopes span −8.3% to +6.0% of the observational slope. Permutation tests (within each model, permuting c_{100} across the dose ladder destroys any dose–response structure) place each family's noise floor at $[-6.9, 6.8]$ (r18), $[-6.9, 5.0]$ (r34), $[-2.5, 43.0]$ (r50), $[-20.2, 21.7]$ (sfn), and $[-2.8, 4.6]$% (regnet). Three of the five shares (ResNet-34, ResNet-50, ShuffleNet) lie inside their floors, so the cross-family sign wobble (ShuffleNet's is negative) is null structure, not a small real channel: a channel carrying a causal signal of the size the observational slope suggests could not flip sign under a family swap while the observational correlation stays at $r \geq 0.87$. Two shares sit marginally *above* their floors' upper bounds (ResNet-18 +7.7% vs. +6.8%; RegNetY +6.0% vs. +4.6%), and the ResNet-50 floor is wide (n=4 slope instability). For these families the null reading does not rest on the floor alone but on the sign-consistent noise harm (8/8 and 4/4 negative) and the near-zero absolute RECOLOR slopes (r18 +0.0105, r50 +0.0017, regnet +0.0068 pp per unit D_{eff}, vs. observational +0.137/+0.087/+0.114). The dissociation is not a ResNet-18 artifact; if anything, the ResNet-18 grid is the *most* generous setting for the geometric path.

8 E6: marker generality

The audit targets one marker. A natural defense of geometry-intervention methods is that D_{eff} is the wrong geometry — participation ratio (PR) or effective rank might be the manipulable one. We therefore recompute two further covariance-spectrum markers on the same ID-test subsample for all 152 E1 cells (8 models × 19 conditions) and ask whether the dissociation anatomy is D_{eff}-specific.

It is not. Both markers anchor observationally ($r(\mathrm{PR}, c_{100}) = +0.77$, $r(\mathrm{erank}, c_{100}) = +0.86$; both track D_{eff} at $r \geq 0.81$). Under ANISO at matched power — the cell where transfer collapses 10.8 pp — PR *rises* by a median +32% and effective rank by +29%: the noise pushes these geometric markers toward the observationally "better" regime while the model gets monotonically worse, the same wrong-signed movement the ISO cell exposes for D_{eff}, now with the harm cell itself. Under RECOLOR(0) both markers move as designed (PR +92% median, erank +144%) with the same ≤ 2 pp transfer neutrality, and ROTATE pins both exactly (relative change +0.000), re-validating the pipeline. The confounded-label reading is thus not an artifact of one marker's definition: every spectrum-based geometry descriptor we tested is a recipe fingerprint, none is a knob.

9 Discussion: geometry as a label

The slope-decomposition arithmetic. Write the observational slope as a sum of channels: $\beta_{\mathrm{obs}} = \beta_{\mathrm{geo}} + \beta_{\mathrm{recipe}} + \beta_{\mathrm{noise}}$, where β_{geo} is the direct causal path from geometry to transfer, β_{recipe} the recipe-level confounding (the recipe sets both), and β_{noise} any channel correlated with geometry change but acting through non-geometric means. Our battery bounds the geometry-only term at a small fraction of the observational slope (point estimate +0.0105, 7.7% of β_{obs}; permutation floor $[-0.009, +0.009]$, with the point estimate at the floor's upper edge; the formation-forcing estimate of §6 lands at 14.7%, placebo-calibrated and paired) and shows that the largest separable non-geometric channel (noise at matched power) moves transfer *against* the direction its D_{eff} change would suggest: $\beta_{\mathrm{noise}} < 0$ while $\Delta D_{\mathrm{eff}} > 0$ (a +5.8 median rise in D_{eff} accompanies a -6.5 pp median transfer drop). What remains of the observational association is recipe confounding. "Intervene on the geometry" methods inherit exactly this arithmetic: no detectable transmission, with a sign-reversing confounder adjacent to the knob.

What the battery does license. Three constructive conclusions survive. *(i) Noise control.* The only large causal effect we can move transfer with is noise, and it moves transfer the right way when removed: adding matched-power noise costs 6–11 pp at pinned or raised D_{eff}, while zero-noise operators of comparable geometric magnitude are transfer-neutral. Spectral-shape-matched noise is the sharpest tool: it is invisible to D_{eff} yet costs 11 pp—any noise-robustness story that monitors only geometry will miss it. *(ii) Formation-time spectral regularization.* Whitening during formation is mildly transfer-positive (+1.7 pp), concentrated in sharper recipes, and the full formation-forcing battery (§6) confirms a real but small channel (14.7% of observational, paired $t = 6.89$) whose dose–response is sub-linear and saturating. This is a representation-shaping regularizer whose value happens to co-occur with larger D_{eff}; studying it as a geometry intervention mistakes the fingerprint for the hand. *(iii) Observational screening.* The correlation itself replicates across architectures ($r \geq 0.87$), so D_{eff} retains value as a cheap *monitoring* signal for recipe drift—it is the interventional reading that fails, not the measurement.

A checklist before reading a marker causally. The battery reduces to five questions any future "geometry intervention" result should answer: (1) *Manipulation check*: does the intervention realize the intended descriptor change ($r \geq 0.99$ here)? (2) *Operator separation*: do two operators that move the descriptor equally but differ in an orthogonal dimension (noise, timing, spectrum) move the outcome equally? (3) *Negative control*: does a descriptor-preserving operator (here ROTATE) leave the outcome unchanged? (4) *Phase separation*: does the effect survive moving the operator from formation to inference? (5) *Forcing saturation*: when the intervention is applied during training at increasing dose, does the outcome scale proportionally, or does it saturate (here: a sub-linear dose–response that flattens to 2% of observational)? A marker whose interventional story survives all five has earned a causal reading; D_{eff} passes none.

Relation to the companion papers. Paper 1 fixed when the observational link holds (ID-matched, architecture-adjusted, shift-typed). Paper 2 fixed what the marker measures (stability map, movement decomposition). Paper 3 (this work) fixes what it does not do: transmit causal intervention. Together they draw the full circle from measurement to mechanism.

10 Limitations

Our conclusions are scoped to: (i) *Scale and dataset*—CIFAR-10 training with CIFAR-100 transfer probes; whether ImageNet-scale recipes exhibit the same recipe-confounding structure is untested, though the mechanism (recipes jointly determine geometry and robustness) has no obvious scale dependence. (ii) *Model count*—8 ResNet-18 recipes chosen to span the widest D_{eff} range the recipe grid allows (2.4×), plus 4 recipes in each of four further architectures (ResNet-34/50, ShuffleNet-V2, RegNetY-400MF); the observational anchor and both separation results replicate in every completed family, but the grid is not exhaustively factorial. (iii) *Transfer readout*—the anchor and E1/E2 use linear probes on frozen features (the protocol that makes the observational anchor clean). The E5d control shows the headline *harm* results are not linear-probe artifacts — the aniso damage *grows* under an MLP readout (−15 to −16 pp) — but the whitening *benefit* is probe-dependent (it vanishes under MLP), and full fine-tuning could behave differently still. (iv) *One layer, one marker family*—all features are penultimate-layer, and although §8 extends the audit to participation ratio and effective rank, all three are functions of the same covariance spectrum; discrete or non-spectral descriptors (e.g. uniformity, alignment) are untested. (v) *Feature-space E3*—formation-phase whitening acts on penultimate features with a refitted head, not on weight-space training dynamics. The E4 forcing battery (§6) closes exactly this gap at scale (96 runs, constraint active at every training step), and the transmission remains at 14.7% of observational with a saturating response surface; the burden of proof now falls on any claim that some other form of weight-space geometric intervention would behave differently.

11 Conclusion

Across the four-part causal battery—including a 96-run formation-phase forcing study, with its registration timeline disclosed in Appendix A—the effective dimension D_{eff} behaves consistently with a readable fingerprint of the training recipe rather than a manipulable cause of OOD generalization: spectral-shape-matched noise collapses transfer monotonically (by up to 11 pp at pinned D_{eff}); moving D_{eff} by up to 8× with zero-noise whitening never moves transfer below baseline (−0.03 to +5.6 pp); the whitening benefit in formation phase decomposes into probe conditioning, while the same operator at inference harms; forcing the geometry *during training* moves D_{eff} to 9.2× baseline at a paired, calibrated slope of 14.7% of observational, with a sub-linear response that saturates at high dose; the entire dissociation replicates across four further architecture families (24 recipes in total); and two further spectrum markers show the same anatomy. The observational correlation—large, stable, and real—transmits to interventions at only a small fraction of its observational size, at the upper edge of the permutation noise floor. Methods that reshape representation geometry to buy robustness should re-derive their mechanisms in these terms: what survives the battery is noise control, not geometry targeting. The broader lesson generalizes beyond D_{eff}: a strong correlation between a representation descriptor and a performance outcome is evidence about *what the recipe did*, and only a battery like this one—operator, phase, and negative controls together—can say whether the descriptor is also something the recipe can usefully be asked to change.

12 Reproducibility statement

All experiments run on public CIFAR data with fixed seeds; preregistration in `paper/adr_0016.md` (Appendices 4–5) written before unblinding; the dose–response falsification criteria (App. 8 + amendment 1) were committed before any E5b datum existed; the E5b adjudication verdict in §5 follows those preregistered branches verbatim. Scripts: `code/intervene_v2_paper.py` (E1–E3), `code/train_forcing_p4.py` + `code/merge_p4_full.py` + `code/dose_shape_fit.py` (E4, 96 runs + dose scan; `merge_p4_full.py` stdout contains the P4-2 arm counts, band exclusions, and per-condition slope medians; the paired t statistics and the net slope $0.0297/12.58 = 0.00236$ (14.7% of 0.0161) are computed from the merged per-run table), `code/intervene_e5_dose.py` (E5b dose–response + MLP control), `code/intervene_e5c_cond.py` (E5c conditioning measurements), `code/intervene_e5d_ceiling.py` and `code/analyze_e5d_ceiling.py` (E5e saturation/headroom test), `code/intervene_e6_marker.py` (E6 second markers), `code/intervene_effect_probe.py`, `code/intervene_recolor.py`; data `runs/{v2_e1_full,v2_e3,v2_e2_r34,v2_e2_r50,v2_e2_sf,v2_e2_reg,v2_e5b,v2_e5d,v2_e6}.json`;

analysis `code/paper3_analysis.py` + `code/paper3_tables.py` + `code/paper3_ci.py` (bootstrap CIs, permutation noise floors, Holm correction) regenerate every number, figure, and table in this paper from the raw JSON (`runs/{...,v2_e5b,v2_e5d,v2_e5c,v2_e6}.json`; and will be released with the code and data archive).

Ethics statement

This work uses public image datasets and does not involve human subjects.

Use of AI tools

Large language models were used as writing and coding assistants. All scientific content, analyses, and reported numbers were determined and verified by the authors.

References

Julius Adebayo, Justin Gilmer, Michael Muelly, Ian J. Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems 31 (NeurIPS)*, 2018.

Alessio Ansuini, Alessandro Laio, Jakob H. Macke, and Davide Zoccolan. Intrinsic dimension of data representations in deep neural networks. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019. arXiv:1905.12784.

Chi-Ning Chou, Hang Le, Yichen Wang, and SueYeon Chung. Feature learning beyond the lazy-rich dichotomy: Insights from representational geometry. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2503.18114.

Chi-Ning Chou, Artem Kirsanov, Yao-Yuan Yang, and SueYeon Chung. Diagnosing generalization failures from representational geometry markers. In *International Conference on Learning Representations (ICLR)*, 2026. arXiv:2603.01879.

SueYeon Chung, Daniel D. Lee, and Haim Sompolinsky. Classification and geometry of general perceptual manifolds. *Physical Review X*, 8(3):031003, 2018.

Michael Cogswell, Faruk Ahmed, Ross Girshick, Larry Zitnick, and Dhruv Batra. Reducing overfitting in deep networks by decorrelating representations. In *International Conference on Learning Representations (ICLR)*, 2016. arXiv:1511.06068.

Amirata Ghorbani, Abubakar Abid, and James Zou. Interpretation of neural networks is fragile. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pp. 3681–3688, 2019. doi: 10.1609/aaai.v33i01.33013681.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pp. 1321–1330, 2017.

Sara Hooker, Dumitru Erhan, Pieter-Jan Kindermans, and Been Kim. A benchmark for interpretability methods in deep neural networks. In *Advances in Neural Information Processing Systems 32 (NeurIPS)*, pp. 9737–9748, 2019.

Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, Quoc V. Le, and Hartwig Adam. Searching for mobilenetv3. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019. arXiv:1905.02244.

Lei Huang, Dawei Yang, Bo Lang, and Jia Deng. Decorrelated batch normalization. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 791–800, 2018.

Lei Huang, Yi Zhou, Li Liu, Fan Zhu, and Ling Shao. Group whitening: Balancing learning efficiency and representational capacity. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 9512–9521, 2021.

Yiding Jiang, Behnam Neyshabur, Hossein Mobahi, Dilip Krishnan, and Samy Bengio. Fantastic generalization measures and where to find them. In *International Conference on Learning Representations (ICLR)*, 2020. arXiv:1912.02178.

Takeru Miyato, Toshiki Kataoka, Masanori Koyama, and Yuichi Yoshida. Spectral normalization for generative adversarial networks. In *International Conference on Learning Representations (ICLR)*, 2018. arXiv:1802.05957.

Preetum Nakkiran, Gal Kaplun, Yamini Bansal, Tristan Yang, Boaz Barak, and Ilya Sutskever. Deep double descent: Where bigger models and more data hurt. In *International Conference on Learning Representations (ICLR)*, 2020. arXiv:1912.02292.

Vardan Papyan, X. Y. Han, and David L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663, 2020. doi: 10.1073/pnas.2015509117.

Ilija Radosavovic, Raj Prateek Kosaraju, Ross Girshick, Kaiming He, and Piotr Dollár. Designing network design spaces. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

Ben Sorscher, Surya Ganguli, and Haim Sompolinsky. Neural representational geometry underlies few-shot concept learning. *Proceedings of the National Academy of Sciences*, 119(43):e2200800119, 2022.

Ranxi Zhou. What does a geometric marker actually measure? a measurement study of out-of-distribution predictors. *Transactions on Machine Learning Research*, 2026a. Under review.

Ranxi Zhou. When do geometric markers predict generalization? a controlled analysis of task-relevance and shift type. *Transactions on Machine Learning Research*, 2026b. Under review.

A Preregistration and deviations

The experiment matrix and pass/fail criteria below were written and committed before any v2 number was analyzed (ADR-0016 App. 4, 2026-11-04 morning; scripts committed alongside). We reproduce the criteria verbatim in substance.

Preregistered matrix. E1: 8 ResNet-18 models × 19 conditions (base; ISO$\in \{0.5, 1, 2\}$; ANISO$\in \{0.5, 1, 2\}$; ROTATE; scale $\in \{0.5, 2\}$; RECOLOR$\in \{1.5, 0.75, 0.5, 0.25, 0\}$; total-spectrum RECOLOR$\in \{0.5, 0\}$; placebo $\in \{1, 2\}$). The ANISO(spectral-shape-matched noise) and ROTATE(spectrum-preserving rotation) cells were preregistered as the two new separation cells. E2: resnet34 / resnet50 / shufflenet_v2_x1_0 / mobilenet_v3_small × lr $\in \{0.5, 0.125\}$ × wd $\in \{2\text{e}{-4}, 5\text{e}{-5}\}$ × seed 0, merged with the E1 ResNet-18s for the cross-architecture anchor. E3: same whitening operator applied (a) at inference vs. (b) during formation on a 60% training subset with a refit head.

Preregistered criteria. *P1*: noise families show per-model $dc_{100}/dD_{\text{eff}} \leq -0.05$ with median ≤ -0.05 and $\geq 6/8$ same sign; zero-noise RECOLOR families |median slope| ≤ 0.02. *P1b*: ANISO at matched power with $|\Delta D_{\text{eff}}|/D_{\text{eff}} \leq 0.2$ yields $\Delta c_{100} < 0$ in $\geq 6/8$ models. *P2*: cross-architecture base $r(D_{\text{eff}}, c_{100}) > 0$ in $\geq 4/5$ architectures, and inference-time RECOLOR slopes $\leq 10\%$ of the observational slope. *P3*: the formation arm lands within ± 0.01 of base c_{100} with ID head intact, while the inference arm degrades by ≥ 0.1 — phases separable.

Deviations. (1) E2 dropped mobilenet_v3_small in favor of regnet_y_400mf: the MobileNetV3 head (Howard et al., 2019) contains a non-linear bottleneck after average-pooling, so penultimate features are not the input space of the linear classifier and feature-space operators would not act on the classifier's input. RegNetY-400MF (Radosavovic et al., 2020) was chosen as a hook-clean lightweight replacement of the same efficiency-oriented family. (2) E3 protocol was revised twice *before any E3 number was analyzed*: a first head-finetuning variant diverged on both arms (train accuracy 0.15–0.36), and a second variant omitted `model.eval()`, letting batch-norm use batch statistics and contaminate round-1 features. The final protocol (head re-initialization + per-dimension standardization + lr 10^{-2}, matching E1's verified probe) was locked before unblinding; both failed variants are archived in the experiment logs. (3) The preregistration anticipated three evaluation GPUs; two additional GPUs became available mid-run and were used only to re-shard the *same* preregistered condition matrix, not to add conditions.

Where the preregistered criteria fail as written. Full honesty requires reporting not only passes. *P1* fails as written: the ISO slope median is -0.008, far above the -0.05 threshold (its D_{eff} movement is huge, so per-unit slopes are tiny), and the ANISO slope is numerically unstable because ANISO pins D_{eff} by construction (division by a near-zero denominator). The paper's noise claim therefore rests on the P1b *endpoint*, which passes cleanly (8/8 negative, Holm $p = 0.012$), not on P1 slopes. *P2*'s share bound passes in every family. *P3* fails as written on both sub-criteria: the formation arm lands at $+0.017$, *outside* the preregistered ± 0.01 window (in the beneficial direction — this is the resurrection channel adjudicated below), and the inference harm, while directionally consistent (7/8 negative, Holm $p = 0.016$), is $30\times$ smaller than the preregistered ≥ 0.1 collapse. We interpret P3's failure as the criterion being mis-calibrated at pre-registration time, not as phase separation being absent; the mechanism adjudication below replaces it.

Amendment 1 (dose–response falsification criteria). Written and timestamped *before any E5b datum existed*, after the E3 endpoint but before the dose–response data. Three mechanism hypotheses for the formation benefit: *H-reg* (generic regularization; predicts dose-flat gains), *H-geom* (representational transmission; predicts monotone gains that persist when the whitened representation is scored in the original space), and *H-cond* (probe conditioning; predicts monotone gains fully reproduced in the probe's own transformed space, absent on original-space readouts). Adjudication rule: the paper's "geometry is not a knob" conclusion is *withdrawn* if and only if arm (b) — whitened-formation representation scored on original features — shows monotone in-dose benefits ≥ 0.5 pp in all models replicating across architectures. *Outcome*: arm (b) is flat-null in every model at every dose; arm (a) reproduces the entire benefit — H-cond confirmed, H-reg and H-geom rejected. The conclusion stands.

Amendment 2 (E4 formation-forcing battery). Preregistered in ADR-0016 App. 16 before any forcing run: criteria P4-1–P4-4 as stated in §6, with a two-stage design (pilot gates → 96-run batch). *Deviations, reported in full*: (1) the forcing loss for the spectral operators was re-implemented mid-pilot after a NaN-poisoning defect (power-iteration autograd blowup); the fix moves the eigenvector solve out of the autograd graph (Rayleigh-quotient gradient, bounded), pilot-batch dispersed/concentrate readings were voided, and the re-run was used throughout—the defect, fix, and voided readings are archived (ADR-0016 App. 18). (2) P4-4's placebo arm triggered its downgrade clause (a placebo moves transfer comparably to low-dose forcing arms); the verdict after placebo-band filtering is unchanged, and the placebo covariance is reported as positive evidence for the confound account. (3) The dose scan (six runs at $\lambda \in \{0.03, 0.3, 3\}$) was added post-batch with its own preregistration (App. 23 of the log): form judged by AICc among linear/power/log, saturation by local-slope decay. (4) One of 96 runs violates the function-preservation rule and is excluded per that rule; all exclusions are listed in the data release.

Amendment 3 (dose-scan seed replication). The App. 23 dose scan characterized response shape on a single seed, so we preregistered a replication before launching any new run (ADR-0016 App. 25): two further seeds × two recipes × three doses (12 runs), with four criteria frozen in advance — R1-s (power-law exponent $b \in [0.3, 0.75]$ and $|\text{median}(b) - b_{s0}| < 0.15$), R2-s (threshold-like cliff shape), R3-s (low-D_{eff} recipe inside the placebo band), and R4-s (function preservation). *Outcome, reported in full*: R3-s passed on both new seeds (all ten points inside the band), R4-s held on 29/30 dose points (the single violation is excluded per rule and listed in the data release), but *R1-s and R2-s failed* — the exponent drifts ($b = 0.502/0.325/0.264$) and the

cliff is absent on the new seeds. The paper therefore states only the cross-seed shape claim ("sub-linear and saturating"; Fig. 4) and reports the failed quantitative criteria as a limitation. Headline paired statistics are unaffected: they are computed over the 96-run batch, not per seed.

B E1 endpoint summary table

Rather than a 152-row dump, we release the full per-cell records as machine-readable JSON (`runs/v2/e1/intervene_v2.json`; 8 models × 19 conditions) and summarize the decisive endpoints here. Table 4 reports per-model endpoint deltas for the three decisive cells of Figure 2.

Table 4: E1 per-model endpoint deltas (pp): ANISO(2), ISO(2), and RECOLOR(0) vs. base. Negative values: transfer loss. The ANISO column is negative in 8/8 models despite $|\Delta D_{\text{eff}}|/D_{\text{eff}} \leq 44\%$; the RECOLOR(0) column is never negative despite D_{eff} multiplying 1–8×. Auto-generated from the locked data.

Model	ANISO(2) Δc_{100}	ISO(2) Δc_{100}	RECOLOR(0) Δc_{100}
`l0.125_w0.0001_s0`	-13.2	-13.1	+2.7
`l0.125_w0.0002_s2`	-10.7	-10.8	+5.6
`l0.125_w2.5e-05_s1`	-13.0	-13.7	+0.2
`l0.5_w0.0001_s1`	-10.8	-5.9	+2.0
`l0.5_w0.0002_s2`	-11.1	-6.0	+1.2
`l0.5_w5e-05_s2`	-9.9	-6.6	+3.1
`l1.0_w2.5e-05_s0`	-10.5	-6.3	+1.4
`l1.0_w5e-05_s0`	-8.6	-4.8	-0.0

C E5d: MLP-probe control

To test whether the E1 findings depend on the linear readout, we repeat the decisive conditions (base, RECOLOR(0), ANISO(2), ROTATE) on two representative models with a 2-hidden-layer (512, 512, ReLU) probe under the identical optimization protocol. The *harm* results strengthen: ANISO(2) costs −15.2 and −16.4 pp under the MLP (vs. −9.9 and −10.7 pp linear) — the noise damage is not an artifact of linear readout. ROTATE remains small (−1.6, −3.9 pp). The whitening *benefit* disappears (+1.1 and −3.3 pp vs. +3.1 and +5.6 pp linear), consistent with the probe-conditioning account of §5: whitening eases *linear* probe optimization specifically. We flag that this control covers two models only; it establishes direction, not precise magnitudes.

D E5e: direct test of the saturation and headroom conjectures

Exploratory (not preregistered; added to convert the two conjectures of the E3 stratification into measurements). We repeat the E5c protocol — conditioning statistics of the probe-train covariance and the 40-epoch shared-protocol probe training curve, in the original versus the γ=0 dose-whitened space — on *all eight* E3 paired models rather than the three E5b models (three of the eight overlap E5c and reproduce its conditioning numbers and final accuracies under seed-pinned reruns; the condition numbers agree exactly in 15/18 conditions and to floating-point tolerance in all 18, the residual differences coming from BLAS-level eigenvalue tail bits).

The two conjectures fare differently.

Operator saturation (supported, sharpened). The model that showed no E3 benefit (lr-1.0/wd-5e−5, baseline $D_{\text{eff}} = 0.87$, the smallest in the grid) has an original-space in-rank condition number of 294 — three to seven orders of magnitude below the other seven models (10^3–10^9). Its covariance is already nearly whitened, so the operator has almost nothing to remove: whitening improves its conditioning by only 22× (median across the other seven: $4.6 \times 10^3\times$), fails to lower its reachable training loss (4.134 → 4.165 at epoch 40; one of three models where the loss does not drop, the other two being slow-converging lr-0.5 probes), and yields the smallest direct probe benefit of the eight (+0.08 pp). Saturation at this recipe is therefore not a property of

the operator but of the representation: the recipe trained a representation whose probe-space spectrum was already flat.

High-D_{eff} headroom (refuted). Across the eight models, baseline D_{eff} correlates *positively* with the direct whitening benefit at γ=0 (Pearson $r = +0.743$, Spearman $\rho = +0.857$, $p = 0.007$): the three high-D_{eff} models ($D_{\text{eff}} \geq 11$) average +3.5 pp of probe benefit against +0.9 pp for the five lower-D_{eff} models. Whatever limits the benefit at the lr-1.0 recipes, it is not exhaustion of the high-D_{eff} models' headroom — recipe sharpness, not geometry, is what correlates with the benefit's absence, consistent with the dose–response adjudication of §5. Data: `runs/v2_e5d_ceiling/intervene_e5d_ceiling.json`.

E E5c: direct conditioning measurements

Exploratory (not preregistered; added after the E5b adjudication to turn the conditioning inference into a measurement). For the three E5b models we train the shared-protocol 100-class linear probe in the original vs. the dose-whitened space (identical 60% training subset, seed-pinned) and record (i) conditioning statistics of the probe-train covariance and (ii) the training curve at epochs $\{1, 2, 4, 8, 16, 24, 32, 40\}$.

Because ReLU features are numerically rank-deficient, the raw $\lambda_{\max}/\lambda_{\min}$ condition number is degenerate; we report the *in-rank* condition number λ_1/λ_r instead (the numerical rank at $\lambda > 10^{-10}\lambda_{\max}$ is itself diagnostic: 494–512 of 512 in the two sharper-recipe ResNet-18 models, but only 99–122 of 512 in the lr-0.5 model, rising to 106–122 after whitening — whitening raises the usable rank there). At γ=0 whitening reduces the in-rank condition number by 10^2–$10^6\times$ (e.g. $2.8\times10^9 \rightarrow 8.4\times10^3$ in `l0.125_w0.0002_s2`), and partial doses interpolate (γ=0.75: 9.3×10^7). Two honest qualifications: first, the conditioning improvement does not track the probe-space benefit monotonically across models (the `l0.5_w5e-05_s2` model shows only a 1.9× reduction yet a +2.1 pp benefit, so the benefit is not a pure condition-number effect); second, whitening *slows early optimization* (epoch-1 training loss 4.05 → 4.50, epoch-1 accuracy 8.8% → 3.7%, consistent with the standardization in the probe protocol partially undoing the whitening early in training) while lowering the *reachable* loss at convergence (epoch-40 training loss 3.43 → 3.01 in the same model, and lower train loss at convergence in 8/9 model–dose pairs; the exception is the smallest benefit model at the $\gamma = -0.5$ dose). The probe-conditioning account of §5 therefore survives the direct measurement in a sharpened form: whitening changes where probe optimization can *end*, not how fast it starts — a property of the measurement space, not of the representation. Data: `runs/v2_e5c.json`.